The empirical recurrence for OEIS A234731, proved by transfer matrix

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Abstract

OEIS A234731 counts 2-column arrays over $\{0,\dots,7\}$ in which every 2×2 subblock has its diagonal sum differing from its antidiagonal sum by exactly 8, and it carries an empirical recurrence of order 7 contributed by R. H. Hardin. It is true, and it is not an empirical matter: the condition constrains only consecutive pairs of rows, so the rows are the vertices of a finite digraph and an $(n+1)$-row array is a walk of length n. Hence $a(n) = \mathbf{1}^\top M^n \mathbf{1}$ on $S = 64$ vertices, which is C-finite. Writing q for the characteristic polynomial of the conjectured recurrence, the residual is $\mathbf{1}^\top M^{n-7} q(M)\mathbf{1}$, so the recurrence holds beyond a threshold exactly when that vanishes from then on — and since the sequence of those values obeys the monic recurrence given by the characteristic polynomial of M, S consecutive zeros settle every later one. The whole test is finite and exact.

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1 The sequence and the conjecture

OEIS A234731 is "Number of (n+1) X (1+1) 0..7 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 8 (constant-stress 1 X 1 tilings).". It has offset 1 and begins

$$168, 672, 2184, 8736, 29976, 119904, 426120, \ \dots$$

The entry carries the following comment:

> Empirical: a(n) = 4*a(n-1) + 34*a(n-2) - 136*a(n-3) - 369*a(n-4) + 1476*a(n-5) + 1260*a(n-6) - 5040*a(n-7).

As of the "Last modified" line on the live entry (Jun 19 2022 01:12:53, revision 10) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 4\,a(n-1)+34\,a(n-2)-136\,a(n-3)-369\,a(n-4)+1476\,a(n-5)+1260\,a(n-6)-5040\,a(n-7) \tag{1}$$

for all $n > 7$, which is every n at which a recurrence of order 7 can be stated.

2 The rows form a digraph

In a 2×2 subblock with top row (r_j, r_{j+1}) and bottom row (s_j, s_{j+1}) the diagonal sum is $r_j + s_{j+1}$ and the antidiagonal sum is $r_{j+1} + s_j$, so the entry's condition on that subblock reads

$$\left| r_j + s_{j+1} - r_{j+1} - s_j \right| \;=\; 8.$$

Every such condition involves two consecutive rows only.

Let V consist of all of $\{0,\dots,7\}^2$, and put $S=|V|=64$. For $r,s\in V$ declare $r\to s$ an edge when $\left|r_j+s_{j+1}-r_{j+1}-s_j\right|=8$ for every j with $1\le j\le 2-1$. Let $M\in\{0,1\}^{S\times S}$ be the adjacency matrix and $\mathbf{1}$ the all-ones vector.

Lemma 1. *The arrays counted by the entry with $n+1$ rows are exactly the walks $r^{(0)}\to\cdots\to r^{(n)}$ of length n in this digraph, so $a(n)=\mathbf{1}^\top M^n\mathbf{1}$.*

Proof. An array with $n+1$ rows and 2 columns is a sequence $r^{(0)},\dots,r^{(n)}$ of rows. Each of its 2×2 subblocks lies in two vertically adjacent rows $r^{(i)},r^{(i+1)}$ and two horizontally adjacent columns $j,j+1$, so the array satisfies the entry's condition on all subblocks precisely when every consecutive pair of rows is an edge; and the row-internal condition, where present, is exactly membership of V. Summing the walk counts $\left(M^n\right)_{r,s}$ over all r and s gives $\mathbf{1}^\top M^n\mathbf{1}$. □

3 The criterion

Let $q(t)=t^7-\sum_i c_i t^{7-i}$ be the characteristic polynomial of (1).

Lemma 2. *For $n\ge 7$, $a(n)-\sum_i c_i a(n-i)=\mathbf{1}^\top M^{n-7}q(M)\mathbf{1}$. Writing $u_j=\mathbf{1}^\top M^j q(M)\mathbf{1}$, the recurrence therefore holds for every $n>t$ exactly when $u_j=0$ for every $j>t-7$. Moreover $(u_j)_{j\ge 0}$ satisfies the monic linear recurrence whose characteristic polynomial is that of M, of degree S; so if u_j vanishes for S consecutive indices it vanishes for every larger index, and the last nonzero u_j determines t exactly.*

Proof. By Lemma 1, $a(n-i)=\mathbf{1}^\top M^{n-i}\mathbf{1}$, so $a(n)-\sum_i c_i a(n-i)=\mathbf{1}^\top M^{n-7}\bigl(M^7-\sum_i c_i M^{7-i}\bigr)\mathbf{1}$, which is the identity. Putting $j=n-7$ gives the equivalence. For the last claim, let χ be the characteristic polynomial of M; by Cayley–Hamilton $\chi(M)=0$, so applying χ to the shift operator annihilates (u_j), and χ is monic of degree S, which is exactly the statement that S consecutive zeros propagate forward. □

4 The computation

Theorem 3. *The recurrence (1) holds for every $n>7$. This is the statement of Section 1.*

Proof. The vector $q(M)\mathbf{1}\in\mathbb{Z}^S$ was formed by 7 matrix–vector products in exact integer arithmetic and the values u_j evaluated for $j=0,1,\dots$. They vanish from $j=7-7+1$ onwards, and the run was continued past S consecutive zeros, which by Lemma 2 settles every larger j. Hence the recurrence holds for every $n>7$. □

5 Verification

Three checks, each independent of the algebra above.

First, the digraph was built directly from the entry's stated condition and the walk counts $\mathbf{1}^\top M^n\mathbf{1}$ compared against the entry's DATA at every one of the 21 published indices; the values agree exactly. That is what ties the digraph to the sequence: a model reproducing only some published terms would be the wrong model and was rejected wherever it occurred.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the vanishing test of Lemma 2 was carried past $S=64$ consecutive zeros rather than to a sampled prefix, in exact integer arithmetic.

References

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